

Zero inflated models

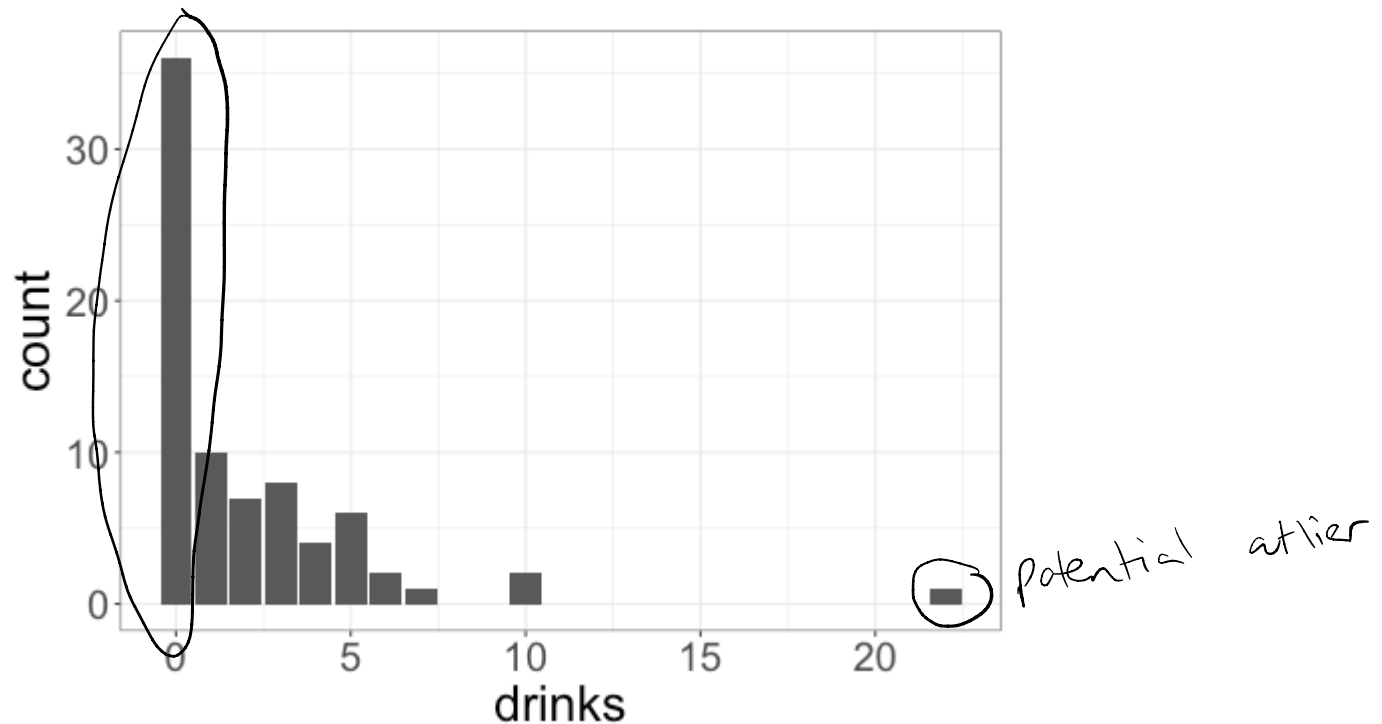
Data: College drinking

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

- + drinks: the number of drinks the student reports consuming
- + sex: an indicator for whether the student identifies as male
- + OffCampus: an indicator for whether the student lives off campus
- + FirstYear: an indicator for whether the student is a first-year student

Our goal: model the number of drinks students report consuming.

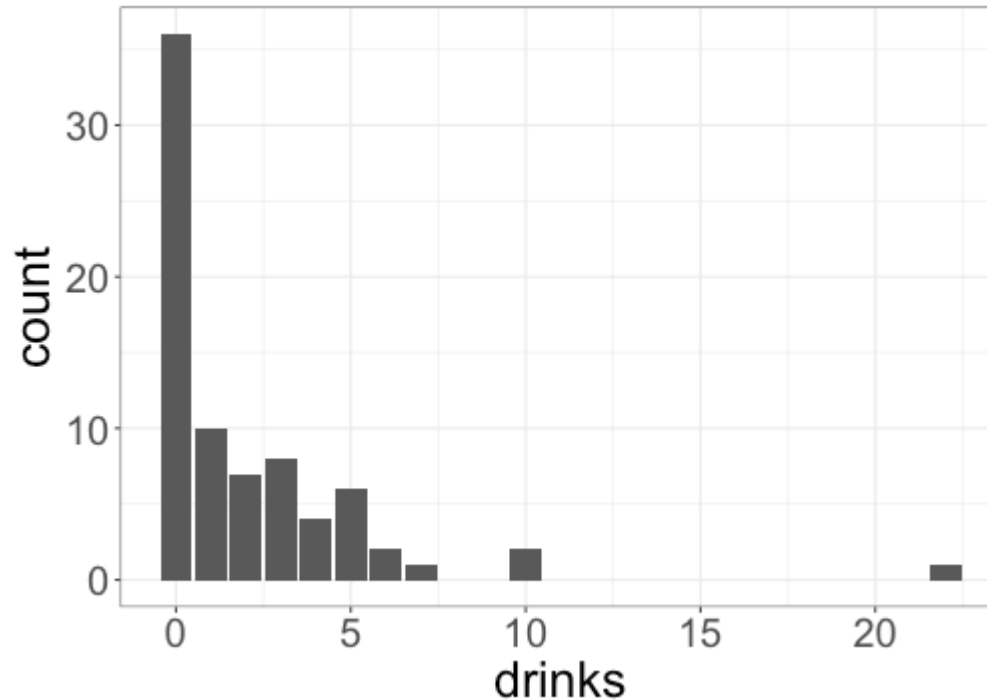
EDA: drinks



What do you notice about this distribution?

- peak at 0
- unimodal, right skewed

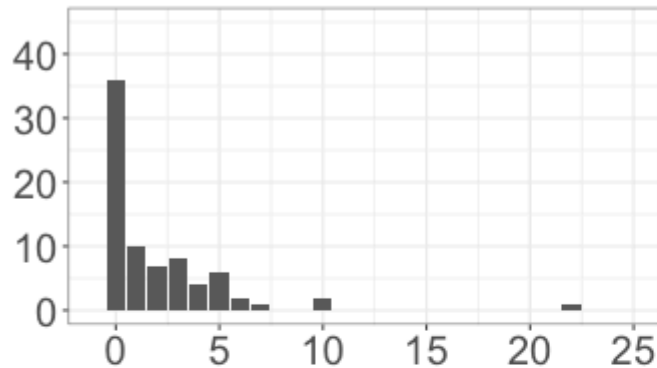
EDA: drinks



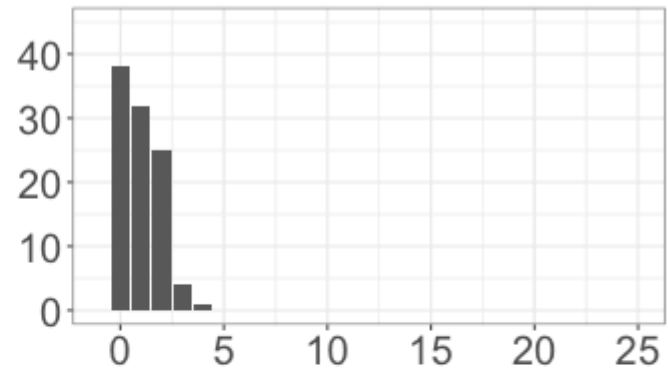
- + The distribution is right skewed and unimodal
- + There is an outlier near 20
- + *There are more zeros than we would expect from a Poisson distribution!*

Comparisons with Poisson distributions

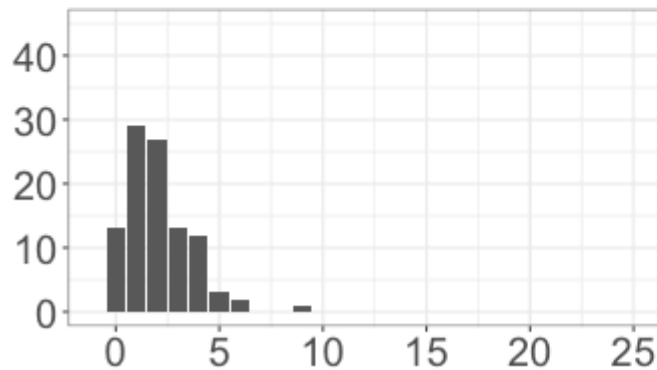
Observed data



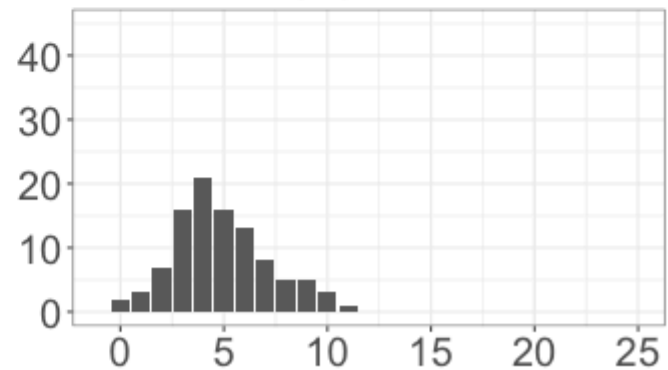
Poisson(1)



Poisson(2)

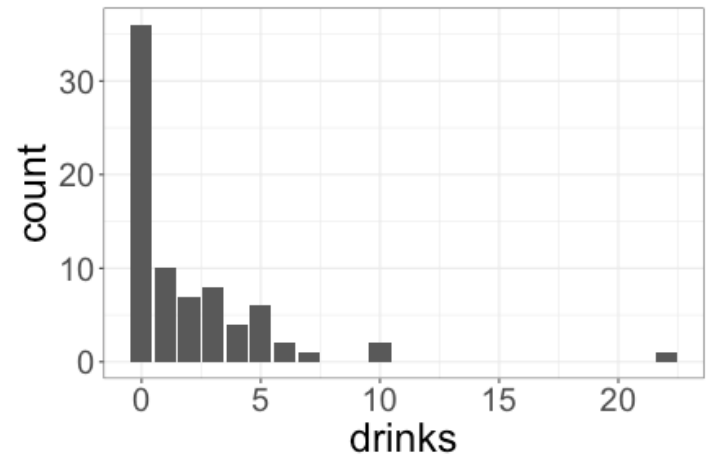


Poisson(5)



Excess zeros

Why might there be excess 0s in the data, and why is that a problem for modeling the number of drinks consumed?



Excess zeros

The problem:

- + There are two groups of people contributing 0s to the data: those who never drink, and those who sometimes drink but didn't drink last weekend
- + By itself, a Poisson distribution doesn't do a good job modeling data that is a mixture of these two groups

Why don't I just include whether or not the student drinks as a variable in the model?

If we observed this variable, we could include it
But we didn't observe it
Plan: create separate models for drinkers &
non-drinkers, then combine

Modeling

Let

- + Z_i denote whether student i is a non-drinker (1 = never drinks, 0 = sometimes drinks) (Z_i is not observed in data,
- + $\alpha_i = P(Z_i = 1)$ but we can imagine trying to model it if it did exist)

We believe that α_i depends on whether or not student i is a first year.

What model can I use for the relationship between being a first year student and being a non-drinker?

logistic regression!

Modeling non-drinkers

Z_i denote whether student i is a non-drinker (1 = never drinks, 0 = sometimes drinks)

$$Z_i \sim \text{Bernoulli}(\alpha_i)$$

$\swarrow$ $P(\text{student } i \text{ is non-drinker})$

$$\log\left(\frac{\alpha_i}{1 - \alpha_i}\right) = \gamma_0 + \gamma_1 \text{FirstYear}_i$$

Modeling drinks

Y_i = number of drinks consumed by student i

If $Z_i = 1$ (the student never drinks), what is the probability of consuming 0 drinks?

$$P(Y_i = 0 \mid Z_i = 1) = 1$$

$$P(Y_i = 1 \mid Z_i = 1) = 0$$

$$P(Y_i = 2 \mid Z_i = 1) = 0$$

$\vdots$

Modeling drinks

- + Y_i = number of drinks consumed by student i
- + Suppose that whether or not a student identifies as male and whether or not a student lives off campus has some relationship with the number of drinks consumed.

If $Z_i = 0$ (the student sometimes drinks), how could I model Y_i ?

if $Z_i = 0$ (sometimes), Y_i (# drinks) is a count variable

$\Rightarrow$ Poisson?

$Y_i | (Z_i = 0) \sim \text{Poisson}(\lambda_i)$

$$\Rightarrow P(Y_i = y | Z_i = 0) = \frac{e^{-\lambda_i} \lambda_i^y}{y!}$$

So far:

Does the student ever drink?

$Z_i \sim \text{Bernoulli}(\alpha_i)$ $\log\left(\frac{\alpha_i}{1 - \alpha_i}\right) = \gamma_0 + \gamma_1 \text{FirstYear}_i$

$P(Y_i = 0 | Z_i = 1) = 1$ $\leftarrow$ If the student never drinks,
 Y_i (# drinks) $\equiv 0$

$Y_i | Z_i = 0 \sim \text{Poisson}(\lambda_i)$ $\leftarrow$ If the student sometimes
drinks, Y_i is Poisson

$\log(\lambda_i) = \beta_0 + \beta_1 \text{OffCampus}_i + \beta_2 \text{Male}_i$

Can we fit these models?

No!, we don't have Z_i

Idea: combine into one model, try to fit
combined model

$$Y_i | (Z_i = 0) \sim \text{Poisson}(\lambda_i)$$

Combining models

(law of total probability)

$$P(Y_i = y) = P(Y_i = y | Z_i = 0)P(Z_i = 0) + \frac{P(Y_i = y | Z_i = 0)}{y!} P(Y_i = y | Z_i = 1)P(Z_i = 1)$$

Case 1: $y = 0$

$$P(Y_i = 0) = \underbrace{P(Y_i = 0 | Z_i = 0)}_{e^{-\lambda_i}} \underbrace{P(Z_i = 0)}_{1 - \alpha_i} + \underbrace{P(Y_i = 0 | Z_i = 1)}_{= 1} \underbrace{P(Z_i = 1)}_{\alpha_i}$$

$$= e^{-\lambda_i} (1 - \alpha_i) + \alpha_i$$

Case 2: $y > 0$:

$$P(Y_i = y) = \underbrace{P(Y_i = y | Z_i = 0)}_{\frac{e^{-\lambda_i} \lambda_i^y}{y!}} \underbrace{P(Z_i = 0)}_{1 - \alpha_i} + \underbrace{P(Y_i = y | Z_i = 1)}_{0} \underbrace{P(Z_i = 1)}_{\alpha_i}$$

$$= \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i)$$

$$y = 1, 2, 3, \dots$$

Zero-inflated Poisson (ZIP) model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases} \quad \text{(random component)}$$

where

systematic component

$$\begin{cases} \log\left(\frac{\alpha_i}{1 - \alpha_i}\right) = \gamma_0 + \gamma_1 \text{FirstYear}_i & \leftarrow \text{logistic piece} \\ \log(\lambda_i) = \beta_0 + \beta_1 \text{OffCampus}_i + \beta_2 \text{Male}_i & \leftarrow \text{poisson piece} \end{cases}$$

This is called a *mixture* model (it is a mixture of two different models). We *can* fit this model on the observed data (we don't need to observe Z_i)

Zero-inflated Poisson (ZIP) model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

where

$$\log\left(\frac{\alpha_i}{1 - \alpha_i}\right) = \gamma_0 + \gamma_1 \text{FirstYear}_i$$

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{OffCampus}_i + \beta_2 \text{Male}_i$$

What do α_i and λ_i represent?

α_i = PL Student i never drinks)

λ_i = average # of drinks if the student does drink

Zero-inflated Poisson (ZIP) model

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What do α_i and λ_i represent?

α_i = probability the student doesn't drink, λ_i = average number of drinks if the student *does* drink

Class activity

Work on the class activity (handout).

Class activity: The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i}\right) = -0.60 + 1.14 \textit{FirstYear}_i$$

$$\log(\hat{\lambda}_i) = 0.75 + 0.42 \textit{OffCampus}_i + 1.02 \textit{Male}_i$$

What is the estimated probability that a first year student never drinks?

The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

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What is the estimated average number of drinks for a male student who lives off campus and sometimes drinks?

The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i}\right) = -0.60 + 1.14 \textit{FirstYear}_i$$

$$\log(\hat{\lambda}_i) = 0.75 + 0.42 \textit{OffCampus}_i + 1.02 \textit{Male}_i$$

What is the estimated probability that a male first year student who lives off campus had at least one drink last weekend?